Proposition I.4
The Side–Angle–Side Congruence Theorem

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1 Introduction

Euclid's fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid's original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid's Proposition I.4, the logical structure of the proof is fundamentally different.

2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

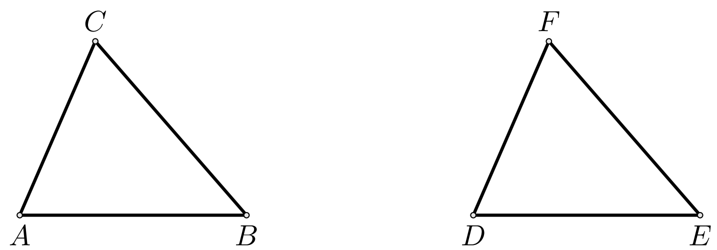
The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

2.1 Final Configuration

The geometric configuration is shown in Figure 1.



Configuration of Proposition I.4.

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

3 Statement

Proposition I.4.

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let ABC and DEF be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4 Proof

Euclid's original proof proceeds by superposition. One triangle is conceived as being placed upon the other in order to establish the coincidence of the corresponding sides and vertices.

The present reconstruction follows a different logical route. Superposition is not part of Hilbert's axiomatic system and therefore does not appear in the formal development.

Instead, the Side–Angle–Side congruence theorem has already been established within the Hilbert theory.

Applying the SAS theorem to the hypotheses

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF$$

immediately yields the congruence of the two triangles.

Consequently,

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This is precisely the conclusion of Proposition I.4.

□

5 Proof Architecture

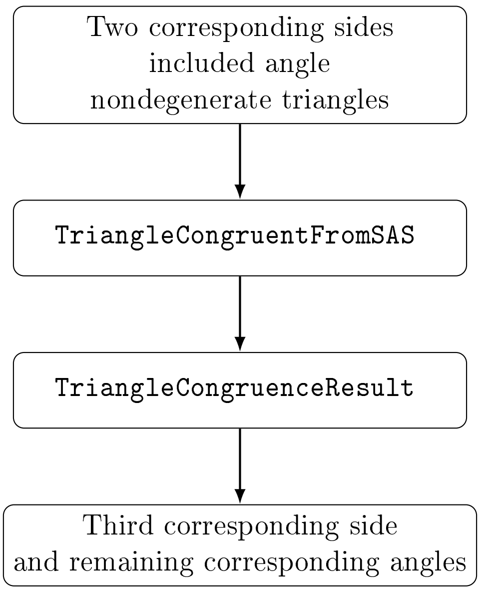


Figure 1: Logical structure of Proposition I.4 in the Hilbert reconstruction.

6 Hilbert Reconstruction

The principal difference between Euclid's original argument and the present reconstruction concerns the method of proof rather than the mathematical statement.

Euclid establishes Proposition I.4 by superposition, treating one triangle as if it could be placed upon the other. Although historically important, this method is not available within the Hilbert axiomatic framework.

Instead, the present development relies on the Side–Angle–Side congruence theorem, which has already been established as part of the Hilbert theory.

Once the hypotheses of the SAS theorem have been established, an application of `TriangleCongruentFromSAS` immediately yields the complete triangle congruence.

Thus the classical proposition is reconstructed without appealing to superposition while preserving exactly the same mathematical content.

7 Lean Formalization

The Lean theorem is a direct application of the previously established Hilbert SAS theorem, exposed through the compatibility theorem `TriangleCongruentFromSAS`.

In addition to the two side congruences and the congruence of the included angles, the formal statement explicitly records that both triples of vertices are noncollinear.

The theorem returns a value of type `TriangleCongruenceResult`. This structure contains the congruence of all three corresponding sides and all three corresponding angles.

Thus the formal proof of Proposition I.4 consists of a single application of `TriangleCongruentFromSAS`; no additional geometric argument is required.

From the viewpoint of formalization, Proposition I.4 is also the first result whose Lean proof is essentially a single application of a previously established interface theorem.